GRAINGUARD · CONCEPT OPTIONS

GrainGuard concept options

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.8 upgrade): the current design as option A and two options with different working principles, each modelled at detail level D3 (kit 1.9); weighted scoring against GGD-URS-001; all three within 10 %, so all go to Amish without a pick

GrainGuard concept options

All three options score within 10 % of each other, so no option is recommended and the choice goes to Amish: option A, the center sensor cable (193 of 245), option C, the fan-side weather controller (192), and option B, the headspace sensor head (186). The close scores hide a sharp split. Option A is the only one that measures the grain itself and meets PR-001 (temperature profile), PR-002 (moisture) and, in part, PR-006 (heating alarm), the needs GrainGuard exists for; option C is less than half the cost and keeps everyone on the ground, but it is a weather-driven fan schedule that never sees the grain. Option A stays the design of record, as the current design of a repo designed before kit 1.8 must (STANDARDS section 21), until Amish decides; options B and C would change the pitch, which is his call in any case. The choice is recorded as "Proposed, awaiting Amish" in GGD-DEC-001 (open item 10) and argued in GGD-DDR-004. Method: STANDARDS section 21.

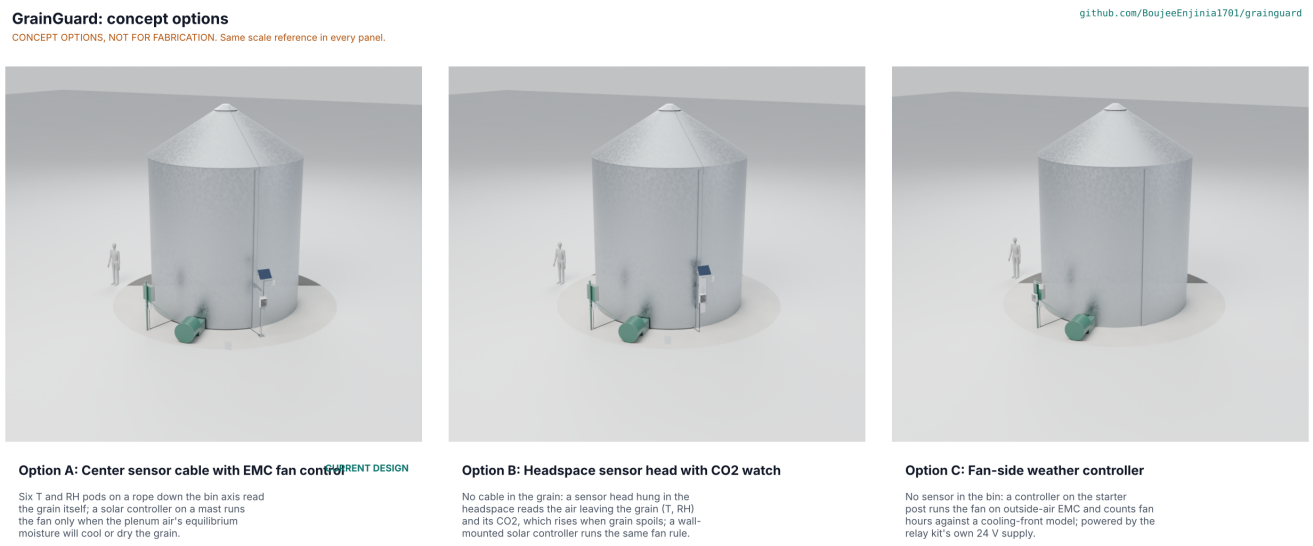


Figure 1. The three options at one scale with a 1.75 m person, rendered from the D3 models in `cad/src/model.py` *(* `components(option)` *) with* `python .kit/options.py` *. The bin, fan and starter are the existing farm equipment of the reference case.*

The options

Option A: center sensor cable with EMC fan control (the current design)

Six temperature and humidity pods on a 6 mm wire rope hang down the bin axis from the bin's center hanger and read the grain itself; a solar 12 V controller on a mast by the bin runs the fan through an electrician-installed relay kit only when the plenum air's equilibrium moisture content (EMC), with the fan heat measured by a plenum probe, will cool or dry the grain. This is the constructable design of GGD-PRC-001, GGD-DDR-003 and the build plan GGD-BLD-001. It measures the temperature profile at 0.94 m pitch and estimates moisture from interstitial RH within 0.23 to 0.42 points from the sensors alone (GGD-CAL-001 B5). Its risks are the moisture estimate's unknown fit error (PR-002), a center cable that watches only 3.3 % of the cross-section (PR-006), the battery in deep cold (STEM check S17) and the roof load during unloading, which depends on the bin maker's hanger rating (PR-010). It needs from a co-design partner a host bin with a rated center hanger, a farmer to set crop targets and an electrician for the relay kit. D3 model: 331 parts (147 fasteners, 19 weld beads), 27.7 kg in `cad/fidelity.json` .

Option B: headspace sensor head with CO_2 watch

No cable in the grain. A printed sensor head, 124 mm across, carrying an SHT45, an NDIR CO_2 module and a small aspirating fan, hangs from the bin's center hanger by an eye bolt in the existing shackle, just under the peak, and reads the air leaving the grain: during a fan run that air carries the top layer's temperature and equilibrium RH, and when grain starts to spoil the headspace CO_2 rises (a Purdue field trial saw 500 to 5,000 ppm as a tank spoiled; STEM check B2). The bus cable leaves through the same peak-cap gland and runs down the wall to a wall unit: a 3 mm aluminium backplate held 0.3 m off the wall stiffener by two welded wall arms on the stiffener's own seam bolts, carrying option A's enclosure, solar panel and radiation shield, so no mast or anchors are needed. Plenum probe, relay kit and conduit are as in option A. It puts almost nothing on the roof hanger, and the head can be reached from the roof with the bin full. It gives no temperature profile and only a bin-top moisture average (B3), its ambient sensor sits close to a sunlit wall, and its wall arms need a calculation and the bin maker's bolt rating (B4). It needs the same partners as A, plus an extension specialist to set CO_2 alarm thresholds for corn. D3 model: 275 parts (150 fasteners, 16 weld beads), 18.9 kg.

Option C: fan-side weather controller

No sensor in the bin. A small IP66 controller on the starter's post, powered from the relay kit's 24 V supply, runs the fan on outside-air EMC (the same rule as A, with the plenum probe measuring the fan heat) and counts fan hours against a cooling-front model (75 to 119 h a cycle; GGD-CAL-001 A8) to judge when a front has passed. The ambient sensor sits in a radiation shield on an arm at the post top. It needs no solar panel, no battery, no mast, no roof work and no bin entry, and it costs less than half of option A. It is a schedule, not a measurement: it cannot see a hot spot, a wet layer or a front that has not passed (C2), so PR-001, PR-002 and PR-006 are not met, and it stops when the starter's supply is isolated. Commercial weather-only aeration controllers and a farmer-built smart-switch system exist (GGD-MKT-001), so its open-hardware value would be cost and documentation. D3 model: 111 parts (45 fasteners), 7.3 kg.

First-order numbers

Table 1. Options compared (costs are estimates for one bin; option A's is the priced BOM).

Quantity	Option A	Option B	Option C
Estimated parts cost (USD)	410.50 (BOM)	about 370 (estimate: A less the rope, pods and mast, plus the sensor head and the wall unit)	about 170 (estimate: controller, ambient shield, relay kit, probe and short conduit; no panel, battery, mast or cable)
Against the USD 260 value-engineering target	USD 150.50 over	about USD 110 over	about USD 90 under
Mass in the D3 model (kg)	27.7	18.9	7.3
What it measures	Grain temperature at 6 levels, interstitial RH (moisture), ambient, plenum	Headspace T, RH and CO_2 (bin-top average), ambient, plenum	Ambient and plenum only
Energy	3.28 Wh per day from a 20 W panel and a 7 Ah AGM battery	About the same plus the head's fan and CO_2 module (to size)	Under 1 W from the starter's 24 V supply
Parts in the D3 model (fasteners)	331 (147)	275 (150)	111 (45)
Install work	Empty-bin entry and roof work for the cable; mast on the pad	Roof work only; wall unit on the stiffener	At the starter post only
STEM check	Within known science; S17 fix proposed	Within known science; B3, B4 plausible	Within known science as a schedule; C2 plausible

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5). UR-013 (cost) is scored on the cost row.

Criterion	Source	Weight	A	B	C	Reason
Grain temperature through the depth	UR-001	3	5	1	1	Only A measures the grain at levels; B reads the headspace, C nothing in the bin
Drying or rewetting	UR-002	3	4	2	1	A estimates moisture at each pod (at risk); B gives a bin-top average after fan runs; C only predicts
Outside air at the bin	UR-003	2	4	3	4	A and C keep the shield in open air; B's shield is 0.3 m from a sunlit wall
Fans run only when they help	UR-004	3	5	4	3	All use plenum EMC; A's cool mode uses measured grain temperature, B the headspace, C a model
Safe when the controller fails	UR-005	3	5	5	5	Same normally open relay, timeout and hand position
Warning of heating	UR-006	3	3	4	1	A sees heating near the center only; B's CO_2 sees spoilage anywhere that vents to the headspace; C cannot
Status from the farmhouse	UR-007	2	5	5	5	Same LoRa link and receiver
No mains, all winter	UR-008	3	4	4	4	A and B solar with the south siting rule; C needs the starter's supply, which every aerated bin has, but stops when it is isolated
Survives the bin's conditions	UR-009	3	3	3	4	A and B carry a lead-acid battery (S17) and electronics in the fumigant; C has neither
No damage to the roof	UR-010	3	3	5	5	A's cable pulls on the hanger during unloading; B's head weighs about a kilogram; C has nothing on the roof
No entry into a bin holding grain	UR-011	3	4	5	5	A is installed with the bin empty and its pods cannot be reached when full; B's head is reached from the roof; C stays at the starter
Mains work kept small	UR-012	3	5	5	4	C also draws its power from the relay kit's supply, so a little more mains-side work
Data stays on the farm	UR-014	2	5	5	5	Same receiver
Retrofit, workshop built	UR-015	2	4	3	5	A has one welded mast; B two welded wall arms and two stiffener seams; C bought parts and one bent arm
Repairable	UR-016	1	4	4	5	All bought modules; C has the fewest parts
Legal radio	UR-017	2	5	5	5	Same radio and packet
Cost against the value-engineering target	UR-013	2	1	2	5	USD 410.50, about USD 370 and about USD 170 against USD 260
Buildable with local skills and tools		2	3	3	5	A and B need welding, printing and wiring; C wiring and one bent arm
Safety		3	3	4	5	A needs empty-bin entry and roof work, B roof work, C ground work only

Criterion	Source	Weight	A	B	C	Reason
Maintenance		1	3	4	5	A's pods are out of reach while the bin is full
Weighted total (of 245)			**193**	**186**	**192**	All within 10 % (B is 3.6 % below A)

Recommendation

No pick: the three options are within 10 % of each other, so all three go to Amish (STANDARDS section 21), recorded as "Proposed, awaiting Amish" in GGD-DEC-001 open item 10 and argued in GGD-DDR-004. Option A remains the design of record until he decides, and the pitch ("Cable of moisture and temperature probes strung through the bin") describes option A only.

The observation for his decision: the weighted sum lets options B and C make up on cost, safety and simplicity what they lose on the three measurements GrainGuard is for. If PR-001, PR-002 and PR-006 are treated as must-haves, only option A passes. If the farmer's first need is cheap, safe fan control and spoilage detection can come later, option C is the cheaper start, and option B is the way to detect spoilage without a cable.

The runner-up ideas worth grafting onto option A, whichever way he decides:

- From option C: where the starter is within reach, power the controller from the relay kit's 24 V supply instead of the panel and battery. That would remove the USD 76 of panel, battery and charge controller, the battery's freezing problem (S17) and its hydrogen (S18), at the cost of a powered cable from the starter to the mast.
- From option B: the headspace CO_2 head as the priced option already allowed for in GGD-DDR-001 (D10), hung from the same shackle above the cable, to see spoilage beyond the center core that one cable cannot watch (PR-006).